

Minas Rolling-Key Protocol (MRP) v1.0: Technical Specification for Secure RC System Telemetry

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1. Introduction and Objectives

Wireless telemetry links in autonomous RC platforms (such as ESP-NOW connections between a transmitter and a base station) are susceptible to packet sniffing, unauthorized command injection, and basic replay attempts ¹. The **Minas Rolling-Key Protocol (MRP)** is a lightweight, bi-directional communication protocol designed for resource-constrained microcontrollers (ESP32 / ESP32-S3) ¹.

The primary objective of MRP is to prevent standard replay attacks and limit the exposure window of encryption keys by rotating them dynamically per successful transmission, while utilizing monotonic counters and a KDF fallback mechanism to handle packet loss.

2. System Architecture and Parameters

- The system comprises two communicating entities:
- Transmitter (T):** The ESP32-S3 microcontroller mounted on the vehicle ¹.
 - Receiver (R):** The base station ESP32 responsible for receiving telemetry and logging data ¹.

Parameter	Symbol	Description
Telemetry Data	D	Struct containing sensor logs, throttle, and steering inputs ¹ .
Monotonic Counters	T_{mc}/R_{mc}	Sequence numbers incremented per packet to prevent reordering and basic replays ¹ .
Active Encryption Key	E_m	The current AES-128 symmetric key used for encryption and

		decryption 1 .
New Encryption Key	E_n	The next key generated by R and transmitted to T in the ACK packet 1 .
Master Seed	S_m	Root secret stored in flash memory, used by the KDF during resynchronization 1 .
Magic Number	Constant	A 32-bit integer (0x4D494E41) appended to payloads to verify successful decryption 1 .

3. Protocol Workflow

Step 1: Telemetry Transmission ($T \rightarrow R$)

1. T increments its counter: $T_{mc} = T_{mc} + 1$ 1 .
2. T packages D , T_{mc} , and the Magic Number, then encrypts them using AES-128 with the active key E_m :

$$\text{Packet} = \text{Enc}(D \parallel T_{mc} \parallel \text{Magic}, E_m)[1]$$

3. T broadcasts the packet via ESP-NOW 1 .

Step 2: Reception and Validation (R)

1. R attempts decryption using the latest expected key E_n , falling back to E_m if decryption fails 1 .
2. Decryption is verified by checking the presence of the Magic Number (0x4D494E41) 1 . If verification fails, the packet is discarded.
3. R checks the monotonic counter gap ($T_{mc} - R_{mc}$). If the gap is greater than 1, missing packets are logged as lost (PacketLost = 1) to preserve dataset integrity 1 .
4. R updates $R_{mc} = T_{mc}$ and logs the data 1 .

Step 3: Key Rotation and Acknowledgment ($R \rightarrow T$)

1. R generates a new key E_n (via RNG or KDF based on S_m and R_{mc}) 1:

$$E_n = \text{KDF}(S_m \parallel R_{mc})[1]$$

2. R encrypts an ACK containing E_n , R_{mc} , and the Magic Number using E_m :

$$\text{ACK} = \text{Enc}(E_n \parallel R_{mc} \parallel \text{Magic}, E_m)[1]$$

3. R transmits the ACK back to T 1.

Step 4: Key Adoption (T)

1. T decrypts the incoming ACK using E_m and verifies the Magic Number 1.
2. T verifies sequence binding by checking if the received R_{mc} matches its current T_{mc} 1.
3. If valid, T adopts the new key: $E_m = E_n$ 1. If mismatched, the key update is ignored 1.

4. Resynchronization Mechanism

To recover from packet loss where ACKs fail to reach T :

- Both T and R maintain a consecutive failure counter (F_c) 1.
- If F_c reaches 5 consecutive failures, both entities compute a resynchronization key independently using the root seed and the last valid counter:

$$E_{\text{sync}} = \text{KDF}(S_m \parallel \text{Last Valid}_{mc})[1]$$

- The system's response to repeated resynchronization triggers (e.g., logging an error, entering a safe mode, or stopping execution) is defined by the implementer according to operational needs 1.

5. Security Limitations and Out-of-Scope Threats

While MRP addresses specific wireless telemetry risks, it has explicit security boundaries and **does not protect** against the following attack vectors:

- **Physical Memory Extraction:** If an attacker gains physical access to the ESP32 and the device lacks Flash Encryption or Secure Boot enabled, the active keys and master seed (S_m) can potentially be dumped from the flash memory.
 - **Denial of Service (DoS) via RF Jamming:** Because ESP-NOW operates on standard 2.4GHz Wi-Fi frequencies, an attacker transmitting continuous broadband noise can disrupt all wireless communication, causing the vehicle to lose connection regardless of encryption.
 - **Side-Channel Attacks:** The current software implementation of AES on standard ESP32 microcontrollers is potentially vulnerable to power analysis (DPA/SPA) or electromagnetic intelligence (EM) monitoring if an attacker has physical probes attached to the chip during operation.
 - **Compromise of the Base Station:** If the base station (R) is physically accessed or infected with malware, the master seed and current encryption keys are exposed.
 - **Replay within a Single Round:** If an attacker records a valid packet and rebroadcasts it before the receiver processes it (though largely mitigated by monotonic counters, raw RF injection can cause transient packet collisions).
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6. References

[1] Eyal Braun. Minas Rolling-Key Protocol (MRP): A Lightweight Secure Telemetry Framework for Autonomous RC Systems. GitHub Repository: EyalBraun/Minas, 2026. [1]